

Assignment on Deep Reinforcement Learning

Amalia Konstantinou

19/5/2025

1 Introduction

The goal of the assignment was to create a program that uses deep reinforcement learning to solve a problem. Deep reinforcement learning is a subfield of machine learning. It is a new method developed the last few years that uses an agent which is being trained. That is achieved by rewarding the correct actions (that lead us towards the goal) and punishes the wrong actions. Also, that is combined with deep learning, such as deep neural networks, which allow us to handle a larger amount and a more complex set of data.

1.1 Project

The project that was chosen was to create a program that handles agriculture production and its goal is to optimize the production of apples. Specifically, an agent learns by interacting with an environment in order to maximize the total reward.

The project focuses on apple production in the island of Crete. Its geographical coverage is longitude: 24.5 - 26.5 and latitude: 34.8 - 35.8. Soil data for that area was used. Namely,

- Clay content.
- Nitrogen content
- The pH of the water
- Sand content
- Silt content

This data was downloaded in the form of TIFF files, which were processed. In the processing stage, also npy files were created, which contain the data in normalized form. These preserve all the data, while allowing faster loading than reading from TIFF files repeatedly. Also, png files were created so as to create a visual representation of the values of these properties across the island. The goal that is set is to reach the optimal conditions for apple production. Which are,

- Clay content [20, 35]

- Nitrogen content [1.5, 3]
- The pH of the water [5.5, 7]
- Sand content [23, 52]
- Silt content [28, 50]

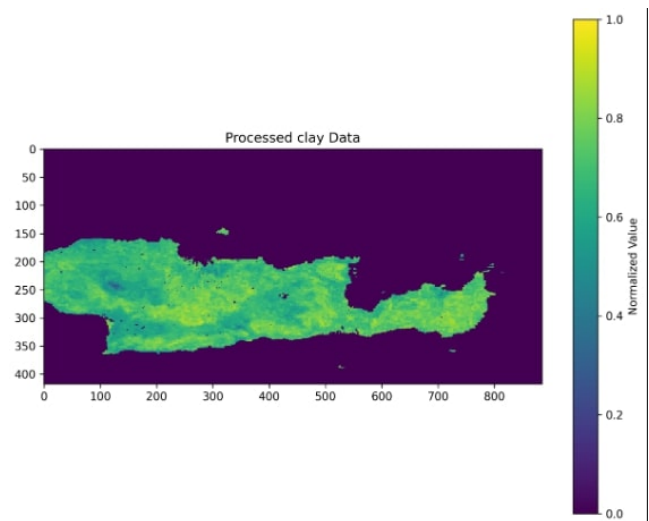


Figure 1. Representation of the distribution of clay in the soil. Layer of 0-5cm below the surface.

1.2 Structure

The project was named `agri_apple_drl`. The main folders of the program are,

- `environment` which contains the environment of the project (`apple_farming_env.py`)
- `configs` with the parameters
- `data` which contains the original unprocessed data, the processed data and the outputs
- `notebooks` with the processed png files
- `utility` with functions, such as `data_processing` and `visualization`
- `main.py` which is the main file running in our project

2 Problem definition

Artificial intelligence has been utilized for the advancement of diseases detection and industrial

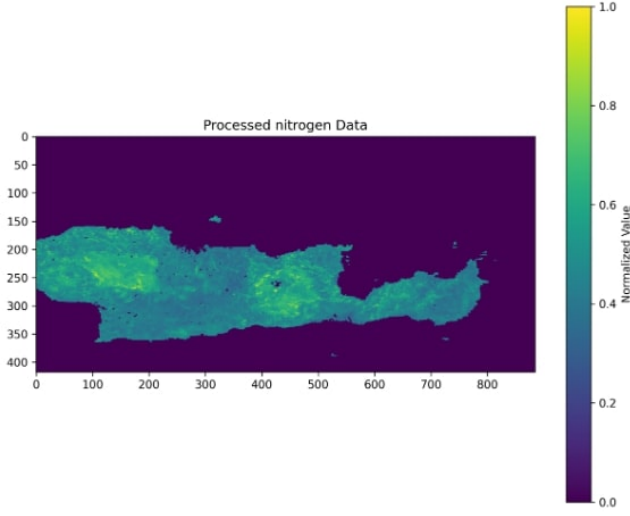


Figure 2. Representation of the distribution of nitrogen in the soil. Layer of 0-5cm below the surface.

robotics. Another field that has seen great improvement thanks to the assistance of AI is agriculture. The positive effect on human life can be astonishing, it will help solve the problem of world hunger by greatly minimizing the cost of food production.

As mentioned above the goal of the project is to optimize the agriculture of apples. Deep reinforcement learning is a very useful tool because it can optimize farming decisions by learning from historical and real-time data, it can adapt to dynamic and constantly changing situations (which is necessary for environmental conditions). So, according to the condition of the soil, it can optimize resource usage - how much to water, how much pesticide to use.

3 DRL Model

3.1 States

At first I used as states only the soil properties - namely clay, nitrogen, pH of water, sand, silt - which are loaded from the file `params.yaml`. Later, I noticed that when adding the last three actions that were taken as states I got more accurate results.

3.2 Actions

The actions that the agent can take are,

- fertilizer
- water
- pesticide

So, the action space is a vector of these three values. The observation space is a Dict that consists of the soil vector, the weather vector, growth stage, the three last actions. Moreover, the growth stages of the apple production were used to create a more realistic representation of the procedure. There are four growth stages - dormant, bloom, development, harvest. These stages have an impact on reward calculations.

3.3 Reward and Penalty

The reward is calculated based on many factors. Firstly, positive rewards are given for reaching better soil quality - it is compared against optimal conditions. Also, for adapting better to the weather conditions, like watering the apples less when it has rained. Also, the reward is affected by the growth stage we are at, each stage has a different multiplier depending on its importance in the growth of apples. For example, the blooming stage is very important so it has a multiplier of 1.1.

3.4 Algorithm

The program contains three different learning algorithms - PPO, A2C, DQN. Each algorithm has its own hyperparameters in the `params.yaml` file. Firstly, in my `SoilFeaturesExtractor` class I defined a neural network, with fully connected layers and ReLU activation functions and layer normalization in order to transform the inputs into a feature representation for the policy network. Then, the algorithms are created inside the `DRLModelFactory` class, inside the `create_model` function. The model is trained. After, the model is evaluated in the class `ModelEvaluator` over many episodes. The function inside it returns: mean reward, standard reward, minimum and maximum rewards, the number and length of episodes.

3.5 Parameters

The hyperparameters for the models were tested and altered. After many tests I reached these values,

- learning rate : 0.00025
- gamma : 0.99
- total timesteps : 500,000

for ppo:

- steps : 2048
- batch size : 128
- epochs : 10

for dqn:

- batch size : 128
- buffer_size: 100000
- learning_starts: 10000
- train_freq: 4

for a2c:

- n_steps: 5
- batch_size: 64

Initially a learning rate of 0.0005 was chosen for faster convergence, but instability in the reward was noticed.

PPO training: I scaled the reward function so that the max reward is 1. The initial reward at the start of training is 0.29. It steadily increases overtime and reaches final values 0.85-0.86, that means the agent is learning and performing well. The kl divergence was a bit high, so I changed the clip range factor to 0.1.

A2C training: The initial reward was 0.303 at the start of training and reaches 0.533 in the end. That is a good number but not ideal. It is evident that PPO is running much better.

DQN training: The DQN algorithm handles discrete actions. So the first thing we have to do is change the action space from Box to Discrete. Specifically, i used Discrete(27). I noticed that it

had the lowest performance, so it was decided not to use it.

4 Improvements

After running the initial code, some errors and miscalculations were noticed. So, some improvements were added to the code.

- some alterations to the parameters were made.
- growing stages were added - dormant, bloom, fruit development, harvest. These create more realistic representation of the way agriculture works, since each stage has different needs and different processes. Each stage has a different weight on the calculation of the reward.
- weather data was added, specifically temperature, rainfall, sunlight. These make the training part more realistic, since the weather greatly affects the growth of apples. So, the actions taken by the agent were centered around the current weather conditions. For example, if it rains less water is added to the crop.
- initially i started with only ppo as a training algorithm. But i wanted to check how different algorithms affect my agent's ability to learn so I added two more. Specifically, A2C which is also an algorithm that is based on continuous actions, and DQN which works with discrete actions.
- also a visualization file was added that depicts the agent's work overtime. It prints diagrams of the actions taken overtime. The visual representation allows us to have a clearer view of the agent's learning ability.
- also I wanted to penalize the excess use of resources since that is costly and has detrimental effect on the plants as well. So, in the calculate reward function i added a penalty for using too much of each resource.

5 Diagrams - Visual Representation

6 Conclusion

Finally, we understand that this is a project with great positive implications on the world. It is

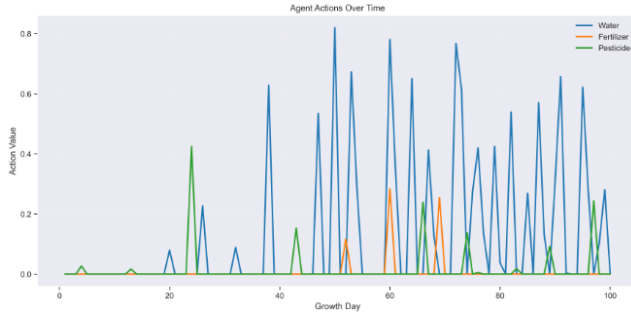


Figure 3. Actions over time.

a complex and big project because of the many parameters you have to take into account (soil, weather, growing stages for plants).

It is apparent that deep reinforcement learning works great with this sector because it is more autonomous and adjusts to the weather and soil conditions. Also, after testing three different algorithms we realized that the most effective one is PPO because it uses continuous actions, which can be calibrated and modified better. Also, it allows us to adjust it a lot thanks to the hyperparameters that are utilized.